

Experiment 4: Crypt-Arithmetic Problem

Aim

To solve a Crypt-Arithmetic problem using Python.

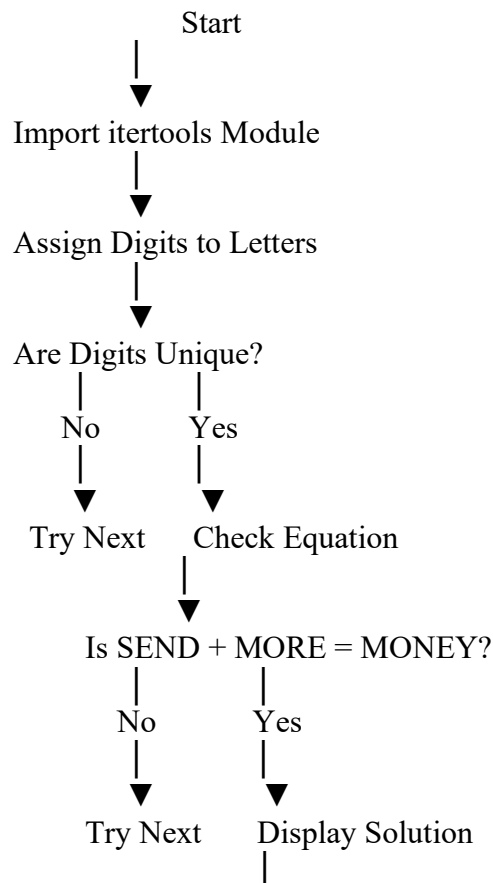
Objective

To find the digit values for letters in a Crypt-Arithmetic puzzle such that the arithmetic equation is satisfied.

Algorithm

1. Import the `itertools` module.
2. Define the words **SEND**, **MORE**, and **MONEY**.
3. Generate all possible digit assignments for the letters.
4. Ensure that no two letters have the same digit.
5. Check if the equation **SEND + MORE = MONEY** is satisfied.
6. If the equation is true, display the solution.
7. Stop the program.

Flowchart



Stop

Python Program

```
from itertools import permutations

letters = ('S','E','N','D','M','O','R','Y')

for p in permutations(range(10), 8):
    S,E,N,D,M,O,R,Y = p

    if S == 0 or M == 0:
        continue

    send = 1000*S + 100*E + 10*N + D
    more = 1000*M + 100*O + 10*R + E
    money = 10000*M + 1000*O + 100*N + 10*E + Y

    if send + more == money:
        print("SEND =", send)
        print("MORE =", more)
        print("MONEY =", money)
        break
```

Sample Output

main.py	Run	Output
<pre>1 from itertools import permutations 2 letters = ('S','E','N','D','M','O','R','Y') 3 for p in permutations(range(10), 8): 4 S,E,N,D,M,O,R,Y = p 5 if S == 0 or M == 0: 6 continue 7 send = 1000*S + 100*E + 10*N + D 8 more = 1000*M + 100*O + 10*R + E 9 money = 10000*M + 1000*O + 100*N + 10*E + Y 10 if send + more == money: 11 print("SEND =", send) 12 print("MORE =", more) 13 print("MONEY =", money) 14 break</pre>		<pre>SEND = 9567 MORE = 1085 MONEY = 10652 === Code Execution Successful ===</pre>

Result

Thus, the Crypt-Arithmetic problem was solved successfully using Python.

Conclusion

The program assigns unique digits to each letter and checks all possible combinations until it finds a valid solution satisfying the equation **SEND + MORE = MONEY**. It demonstrates the use of **permutations** and **constraint satisfaction** in Artificial Intelligence.

